

What a Reference Model and Boreholes Add to a Tuned Lp Magnetic Inversion

An airborne magnetic survey over two buried bodies, inverted with Lp (sparse) regularization under three kinds of prior knowledge: none, an interpreted geological model with realistic errors, and five boreholes. The true model is known, so every inversion can be scored. Each of 40 regularization settings was run with each prior (200 inversions), which separates what the prior does from what the settings do.

MODEL

5 × 5 km (UTM 43N). A cube, 300 m, 100–400 m below the ground, 0.5 SI; an intrusion 250 m thick and 1.2 km long, 100–600 m deep, dipping 45° east, 1.0 SI

DATA

Total-field anomaly 80 m above the ground on a 50 m grid (10,201 points, -2,868 to 3,057 nT), noise 2 % + 5 nT; inverted at 100 m (2,601 points). Inducing field of the Karnataka survey area: inclination 19.24°, declination -1.32°

INVERSION

OcTree mesh, 50 × 50 × 25 m core cells, 233,268 active cells; susceptibility 0–3 SI; Lp regularization with depth weighting; up to 200 iterations (100 IRLS)

RUNS

200 inversions on 10 AWS c5.4xlarge instances (ap-south-1), four at a time on each: 102 min, about \$12; 6–22 min and 19–108 iterations each

Key findings

- **The regularization settings matter as much as any prior.** With no prior, the 40 settings score from 0.23 to 0.80 (median 0.58) on the same data, all fitting it equally well. The smoothness length scale matters most ($L = 3$ averages 0.66, $L = 1$ 0.42); the norms (1,1,1,1) are the most robust (0.72 on average); depth weighting $\beta = 3$ is worse. GeoInv3D's former default Lp setting scored 0.31; it now uses $p = (1,1,1,1)$ at $L = 3$ for magnetic data, which scores 0.74 without a prior and 0.76 with the holes.
- **Well tuned, the magnetic data alone recover both bodies.** The best setting without a prior ($p = (1,1,1,1)$, $\beta = 1.5$, $L = 1$) scores 0.80: the cube's magnetization is centred at 242 m (true 262 m), the intrusion dips 55° (true 45°) and holds 0.79 SI on average (true 1 SI).
- **Boreholes are the prior that helps, and they help most where the inversion is weakest.**
 - With five holes reaching 150 m, the best setting scores 0.87, and the dip comes to 47°.
 - The holes raise the score at 35 of the 40 settings: by a median of +0.21 where the inversion alone scores below 0.5, and +0.00 where it already scores 0.7 or more.
 - The worst quarter of settings with holes (0.58) scores as well as the median without them (0.58); 18 settings reach 0.7 with holes, 10 without.
 - The holes' effect reaches about 250 m, and a test hole 224 m from any of them sees the intrusion's susceptibility recovered (0.98 SI against 1.16 without; true 1).
- **An interpreted model about 100 m out does not help, and hurts more the harder it is enforced.** As a reference model at weight 1, 10 and 100 it scores at best 0.81, 0.70 and 0.53 (no prior: 0.80), and beats no

prior at the same setting 25, 15 and 10 times out of 40. It leaves magnetization where it was drawn wrongly (0.38–0.47 SI against 0.29) at the same data fit, so the misfit gives no warning.

- **Recommendations:** use a smoother length scale ($L = 3$) and the norms (1,1,1,1), now GeoInv3D's default for magnetic data; bring boreholes in with a radius of influence; give interpreted bodies a weight of 1 at most and always compare with the inversion without them.

1 The question

An inversion of magnetic data needs two kinds of decisions: how to regularize it (the norms that make the model compact or smooth, the depth weighting, the smoothness) and what prior knowledge to put in. Both are uncertain in practice, and the data cannot settle either: many different models fit them equally well. This study asks, on a model whose answer is known:

- how much the regularization settings alone change the result;
- whether an interpreted geological model, roughly right but not exact, improves the result when it is used as the reference model, and how strongly it should be weighted;
- whether a few boreholes improve it, and whether they do so across settings or only for a lucky one.

2 The model and the data

Two magnetic bodies lie under random hilly ground (510–750 m above sea level) in a 5×5 km area: a cube, 300 m on a side, 100–400 m below the ground at 0.5 SI, and a stubby intrusion, 250 m thick, 1.2 km long and 100–600 m deep, dipping 45° east at 1.0 SI (Figure 1). The intrusion has about 8 times the cube's volume. The inducing field is that of the Karnataka survey area: at an inclination of 19.24° the anomaly of each body is mainly a low over it with a high to the south. The data were computed on 25 m cells, a different mesh from the inversion's, with noise of 2 % plus 5 nT.

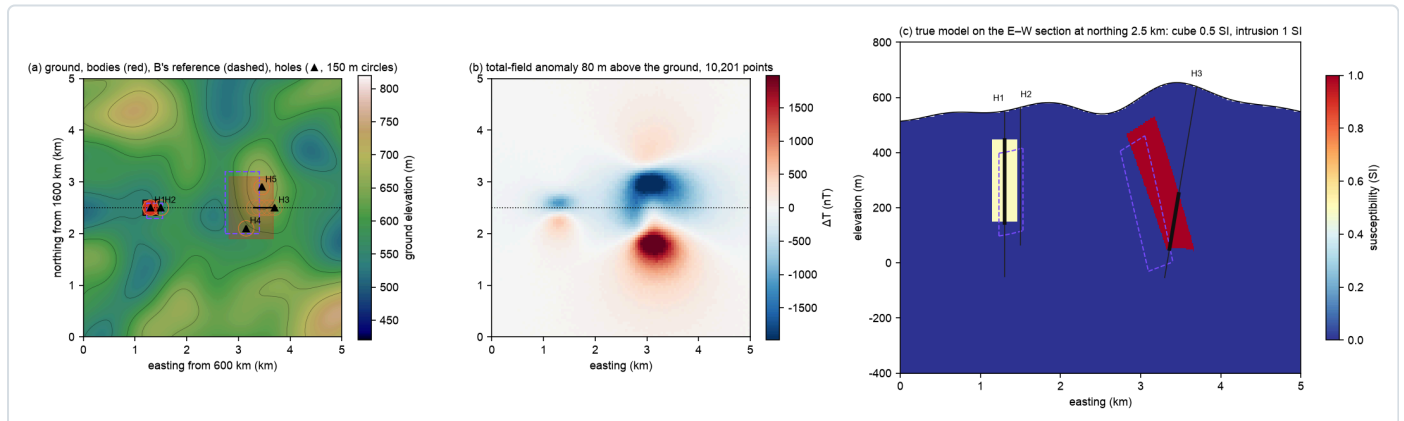


Figure 1 The synthetic model. (a) The ground, the true bodies (red: the cube, and the intrusion's footprint as it dips), the interpreted bodies of group B (dashed) and the holes of group C (triangles, with 150 m circles along them); dotted: the section shown in later figures. (b) The total-field anomaly with noise. (c) The true model on the section, with group B's interpretation (dashed) and the holes (thick where they cross a body).

3 The inversions

Every inversion uses L_p (sparse) regularization, solved by iteratively reweighted least squares (IRLS) until the data are fitted to their noise ($\chi^2/N = 1$). Three of its settings were varied, 40 combinations in all:

- **norms** $p = (p_s, p_x, p_y, p_z)$: (0,0,0,0), (0,1,1,1), (0,2,2,1), (0,2,2,2), (1,1,1,1). p_s acts on the model, $p_{x,y,z}$ on its gradients: 2 favours smooth, 1 blocky and 0 compact models.

- **depth weighting** β : 1, 1.5, 2, 3. The cells are weighted by $(z + z_0)^{-\beta/2}$; a larger β makes deep magnetization cheaper.
- **length scale** $L = \alpha_x = \alpha_y = \alpha_z$ (SimPEG's length scales): 1, 3. A larger L weighs the gradients more against the model itself: smoother models.

The rest is fixed: $\alpha_s = 1$, the first trade-off parameter from the largest eigenvalue, halved each iteration, automatic IRLS thresholds, at most 200 iterations of which 100 IRLS. Each setting was run in five groups (Table 1).

Table 1. The five groups

group	prior	cells constrained (over half / in part)
A: no prior	reference model 0 everywhere; bounds 0–3 SI	—
B: reference model, weight 1	the interpreted bodies of Table 2 as the reference model, at weight 1; bounds 0–3 SI	994 (+340 in part)
B: reference model, weight 10	the interpreted bodies of Table 2 as the reference model, at weight 10; bounds 0–3 SI	994 (+340 in part)
B: reference model, weight 100	the interpreted bodies of Table 2 as the reference model, at weight 100; bounds 0–3 SI	994 (+340 in part)
C: boreholes, 150 m reach	five logged holes (Table 3): ± 0.005 SI and weight 5 on the trace, fading linearly to nothing at 150 m	316 (+1,306 in part)

Group B stands for a geologist's interpretation that is roughly right: each body about 100 m out of place, 50 m too deep, the intrusion 10° too steep, and the susceptibilities 50 % off (Table 2). It enters as the reference model only: the inversion is pulled towards it, more strongly the larger its weight (1 = as strongly as every other cell is pulled towards zero), but its bounds are those of every cell, 0–3 SI, so the data may change it.

Table 2. Group B's interpretation against the true model

	true model	interpreted (group B)	error
cube centre (E, N)	601,300, 1,602,500	601,380, 1,602,440	+80 m E, -60 m N
cube depth	100–400 m	150–450 m	50 m deeper
cube susceptibility	0.5 SI	0.75 SI	+50 %
intrusion's top (easting)	603,000	602,900	-100 m
intrusion's strike (northing)	1,601,900–1,603,100	1,602,000–1,603,200	+100 m
intrusion depth	100–600 m	150–650 m	50 m deeper
intrusion dip	45° E	55° E	10° steeper
intrusion susceptibility	1.0 SI	0.5 SI	-50 %

Group C has five holes logging the true susceptibility (Table 3). The cells a hole passes through are held at the log (± 0.005 SI, weight 5); the cells around it share in the log, the share falling linearly from 1 at the hole to 0 at 150 m (the model builder's `radius_m`): a cell 75 m away has a reference, bounds and weight half the log's and half the unconstrained ones. The intervals logged as background (0 SI) reach out in the same way.

Table 3. Group C's holes (logs from the true model)

hole	collar (E, N)	direction	length	magnetic intervals (along the hole)
H1, through the cube's centre	601,300, 1,602,500	vertical	600 m	105–405 m: 0.5 SI
H2, 50 m east of the cube	601,500, 1,602,500	vertical	500 m	background (0 SI) throughout
H3, inclined, through the intrusion	603,700, 1,602,500	azimuth 270°, 60° down	800 m	450–680 m: 1 SI
H4, the intrusion's shallow part	603,150, 1,602,100	vertical	500 m	100–430 m: 1 SI

hole	collar (E, N)	direction	length	magnetic intervals (along the hole)
H5, the intrusion's deeper part	603,450, 1,602,900	vertical	800 m	375–605 m: 1 SI

4 How the results are scored

The main measure is the **volume-matched overlap**. In the cube's half of the area, take the cells of highest susceptibility until they add up to the cube's volume, and ask what share of the cube they cover; do the same for the intrusion in its half. The **score** is the mean of the two: 1 is a perfect recovery. It judges position and shape only, not amplitude, so it is fair both to compact models (a little volume at high susceptibility) and to smooth ones (a lot of volume at low).

Further measures, all compared with the true model measured the same way: the depth of the centre of the cube's magnetization (true 262 m), the depth above which 90 % of the intrusion's magnetization lies (true 550 m), the intrusion's apparent dip (true 45°), the mean susceptibility in each body, and, for group B, the mean susceptibility where its interpretation is wrong (cells of the interpreted bodies with no true body).

Choosing the best setting by its score needs the true model, which real surveys lack: the best results here are what tuning could achieve at most. The median over the 40 settings stands for a setting chosen without that knowledge.

5 How much the settings matter

All 200 inversions fit the data equally well (χ^2/N 0.85–1.12; 199 reached the target within the iteration limit), yet their scores differ widely (Figure 2, Table 4). Three patterns hold (Figure 3, Table 5):

- **The length scale matters most, and $L = 3$ is better in every group.** At $L = 1$ the gradients weigh too little, and gradient norms of 2 squeeze the magnetization into thin sheets: without a prior, $p = (0,2,2,2)$ scores 0.25 at $L = 1$ and 0.76 at $L = 3$ ($\beta = 1.5$).
- **The norms $(1,1,1,1)$ average highest in every group** (0.72 without a prior) and depend least on the other settings. $(0,0,0,0)$ varies more, but gives the best result with the holes; $(0,2,2,1)$ and $(0,2,2,2)$ average lowest in most groups.
- **Depth weighting between $\beta = 1$ and 2 changes little;** $\beta = 3$ puts the bodies too deep and lowers the score without a prior (0.48 on average against 0.57 at $\beta = 1.5$), with the reference model at weight 1 and with the holes. When the reference model is weighted 10 or 100 it dominates, and β matters little.

GeoInv3D's former default L_p setting, $p = (0,2,2,1)$ with $\alpha = 1$ ($L = 1$) and $\beta = 1.5$, scored 0.31 without a prior, one of the weaker settings here. Following this study, a single magnetic inversion now defaults to $p = (1,1,1,1)$ at $L = 3$ with $\beta = 1.5$ (0.74 without a prior, 0.76 with the holes); the other methods keep the former default, as they were not tested.

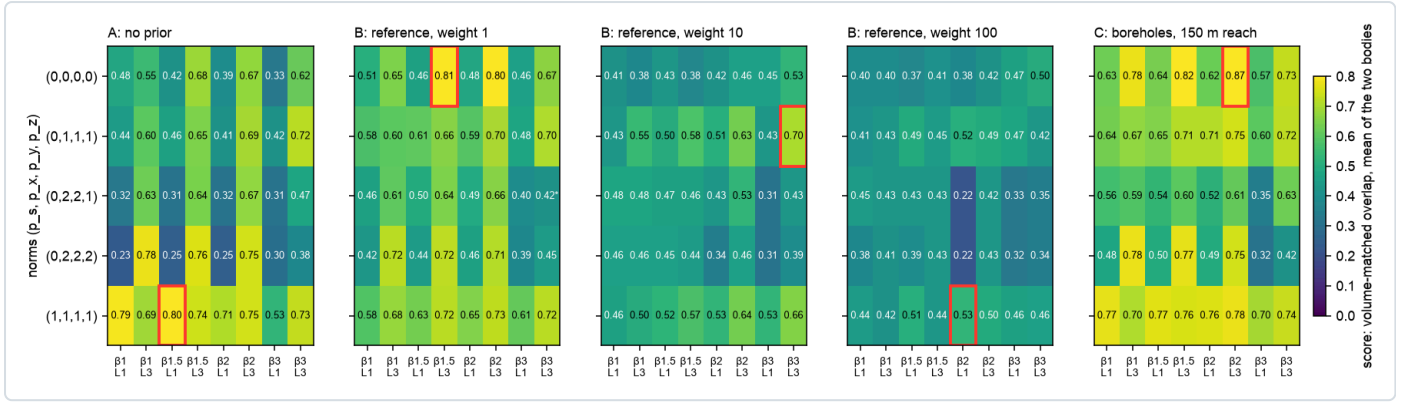


Figure 2 The score of every inversion: rows are the norms, columns the depth weighting β and the length scale L ; * not converged within the iteration limit; red: the best of each group.

Table 4. The scores of each group over the 40 settings

group	best	median	middle half	worst	beats A at the same setting	converged
A: no prior	0.80	0.58	0.39–0.70	0.23	—	40 / 40
B: reference model, weight 1	0.81	0.61	0.47–0.68	0.39	25 / 40	39 / 40
B: reference model, weight 10	0.70	0.46	0.43–0.53	0.31	15 / 40	40 / 40
B: reference model, weight 100	0.53	0.43	0.40–0.46	0.22	10 / 40	40 / 40
C: boreholes, 150 m reach	0.87	0.66	0.58–0.75	0.32	35 / 40	40 / 40

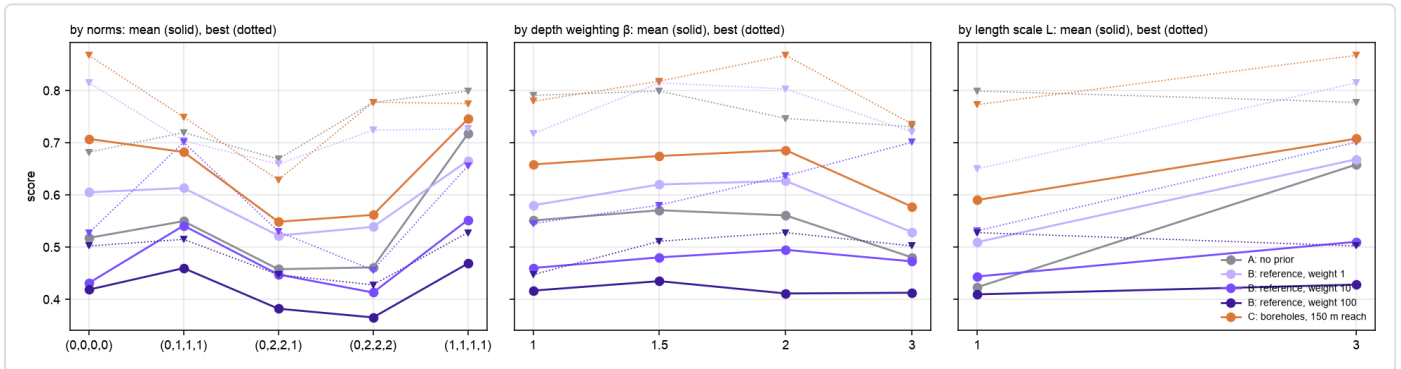


Figure 3 The mean score (solid) and the best (dotted) at each value of one setting, over the other settings.

Table 5. The mean score at each value of one setting

setting (averaged over the others)	A	B, w 1	B, w 10	B, w 100	C
$p = (0,0,0,0)$	0.52	0.61	0.43	0.42	0.71
$p = (0,1,1,1)$	0.55	0.61	0.54	0.46	0.68
$p = (0,2,2,1)$	0.46	0.52	0.45	0.38	0.55
$p = (0,2,2,2)$	0.46	0.54	0.41	0.37	0.56
$p = (1,1,1,1)$	0.72	0.67	0.55	0.47	0.75
$\beta = 1$	0.55	0.58	0.46	0.42	0.66
$\beta = 1.5$	0.57	0.62	0.48	0.44	0.67
$\beta = 2$	0.56	0.63	0.50	0.41	0.69
$\beta = 3$	0.48	0.53	0.47	0.41	0.58
$L = 1$	0.42	0.51	0.44	0.41	0.59
$L = 3$	0.66	0.67	0.51	0.43	0.71

6 The best result of each group

Table 6. The best setting of each group. Depths in m: the centre of the cube's magnetization, and the depth above which 90 % of the intrusion's lies; dip of the intrusion in degrees; κ: mean susceptibility (SI) in each body, and where group B's interpretation is wrong. The first row is the true model, measured the same way

group, best setting	χ^2/N	score	overlap: cube	overlap: intrusion	cube depth	intrusion depth	dip	κ cube	κ intrusion	κ wrong
true model, measured the same way	—	1.00	1.00	1.00	262	550	45	0.50	1.00	—
A: no prior p (1,1,1,1) · β 1.5 · L 1	0.94	0.80	0.83	0.77	242	579	55	0.30	0.79	0.24
B: reference model, weight 1 p (0,0,0,0) · β 1.5 · L 3	0.98	0.81	0.83	0.80	240	562	57	0.26	0.62	0.31
B: reference model, weight 10 p (0,1,1,1) · β 3 · L 3	0.97	0.70	0.77	0.63	292	857	60	0.36	0.60	0.42
B: reference model, weight 100 p (1,1,1,1) · β 2 · L 1	0.99	0.53	0.53	0.53	281	613	57	0.36	0.57	0.47
C: boreholes, 150 m reach p (0,0,0,0) · β 2 · L 3	0.91	0.87	0.87	0.87	240	552	47	0.32	0.89	0.09

Without a prior, the best setting places the cube's magnetization at 242 m (true 262 m) and 90 % of the intrusion's above 579 m (true 550 m), but makes the intrusion about 10° too steep (55°). With the holes the dip comes to 47° and the susceptibility in the intrusion to 0.89 SI, the closest of all. Each of group B's best results keeps part of the magnetization where the interpretation put it (Figure 4): the cube east of and below its true place, the intrusion west of it and steeper. The heavier the weight, the more stays there, and the less the true intrusion holds (0.62, 0.60 and 0.57 SI).

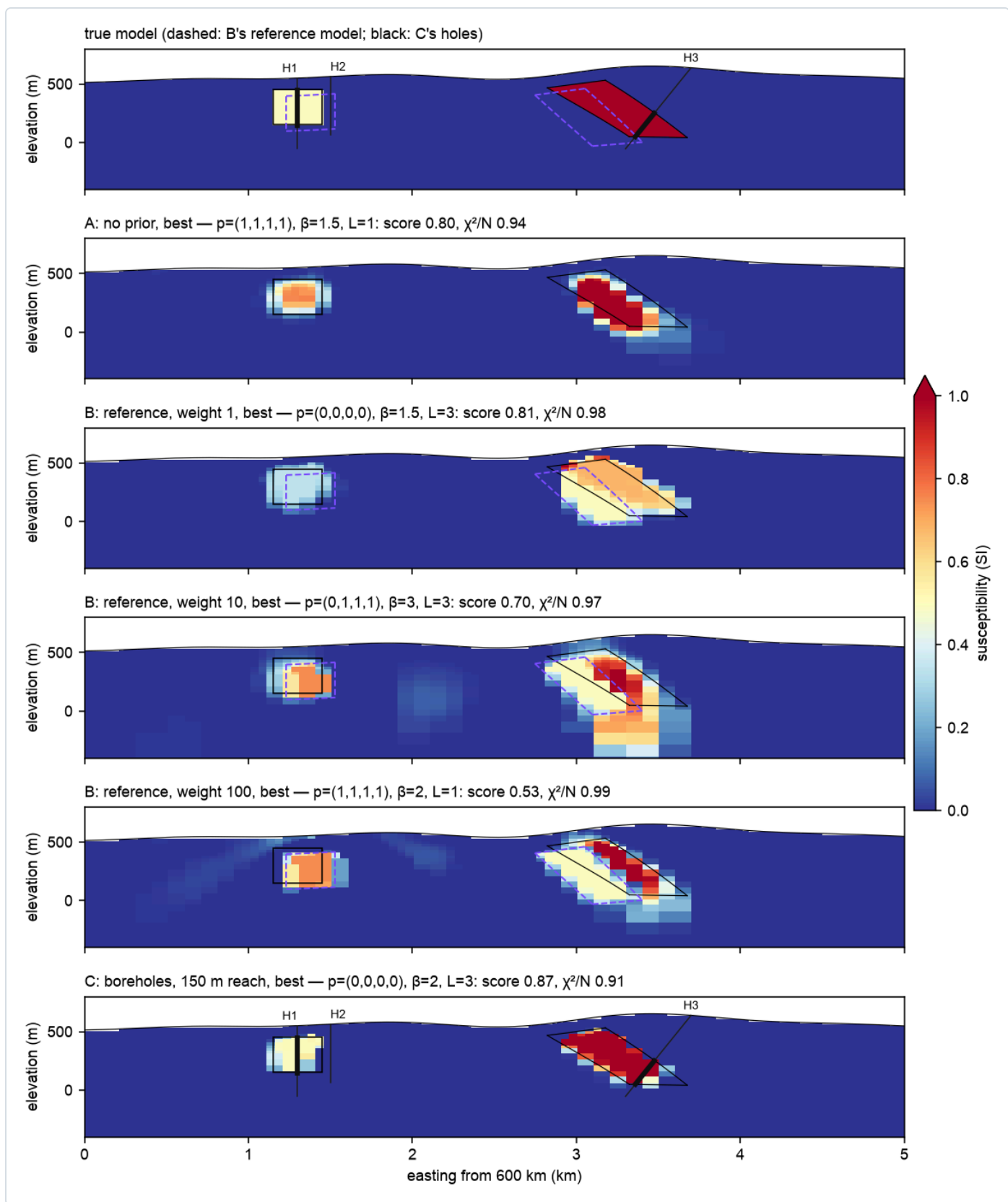


Figure 4 East-west sections (northing 2.5 km) of the best setting of each group. Black outlines: the true bodies; dashed: group B's interpretation; thick black: the holes' magnetic intervals. Colour scale 0–1 SI.

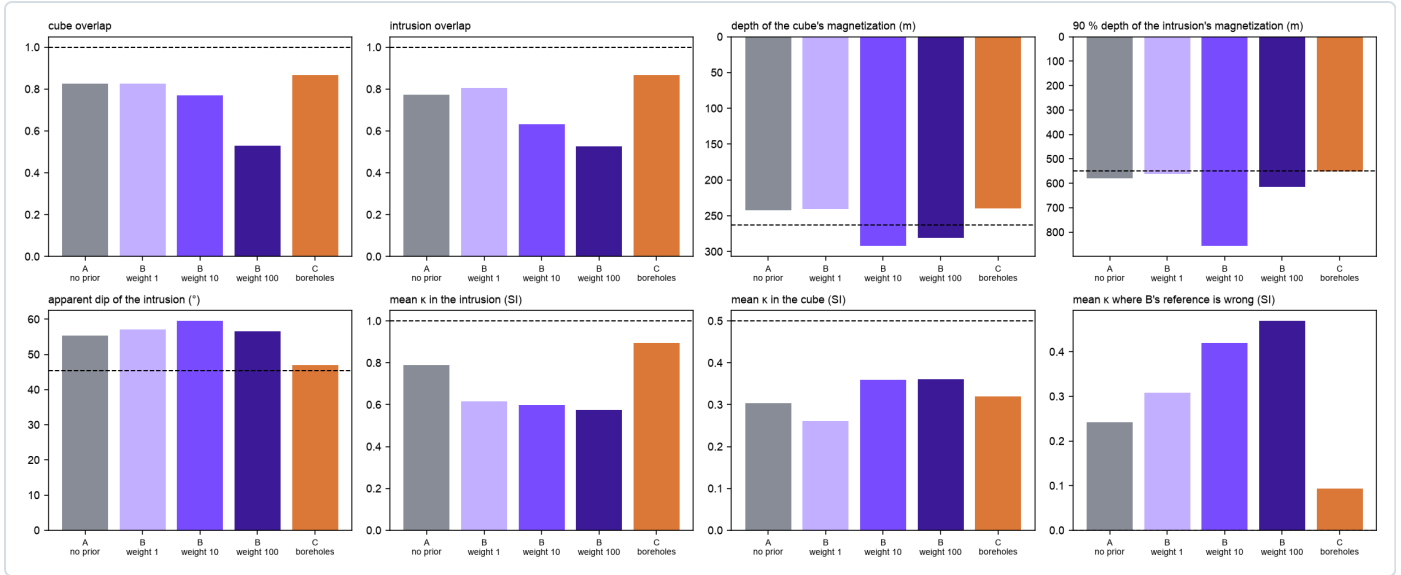


Figure 5 The measures of each group's best result; dashed: the true model measured the same way. Depth axes point down.

7 Boreholes: better across the settings

A best result can be luck: one setting out of 40 that happens to suit the model. What matters in practice is whether the holes help whatever setting is chosen. Run with and without the holes at each of the 40 settings (Figure 6):

- **The holes raise the score at 35 of the 40 settings.** The median rises from 0.58 to 0.66, the worst from 0.23 to 0.32, and the worst quarter of settings with holes (0.58) does as well as the median without them.
- **They help most where the inversion alone is weakest.** Where it scores below 0.5 (18 settings), the holes add a median +0.21; where it already scores 0.7 or more (10 settings), +0.00. The largest gains are with the most compact norms, (0,0,0,0) (+0.12 to +0.24), and with gradient norms of 2 at $L = 1$ (+0.01 to +0.25), the settings that otherwise squeeze the bodies into thin sheets (Figure 8). With the robust norms (1,1,1,1) the change is small (-0.03 to +0.05 at $\beta \leq 2$).
- **The losses are small.** The largest is -0.06 ($p = (0,2,2,1)$, $\beta = 2$, $L = 3$); only 5 settings lose at all.

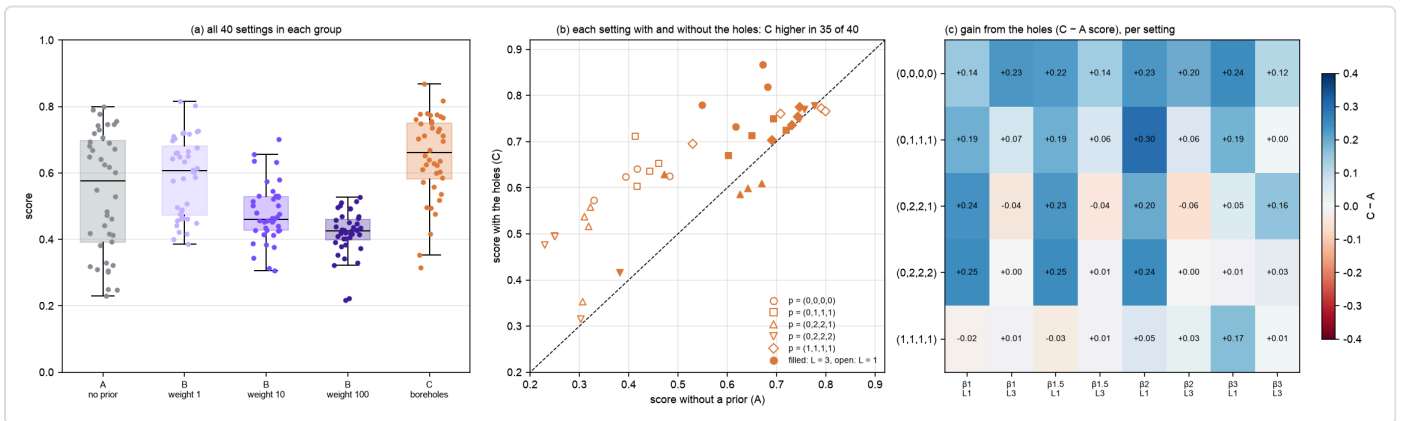


Figure 6 (a) The scores of all 40 settings in each group (boxes: the middle half; whiskers: the range). (b) Each setting without (horizontal) and with the holes (vertical): above the dashed line the holes help. (c) The change in score from the holes at each setting.

Measure by measure (Figure 7, Table 7), the holes narrow the spread most for the cube, which hole H1 passes through: its overlap falls below 0.4 at 2 settings with the holes against 14 without, and the middle half of the settings ranges 0.64–0.81 against 0.33–0.77 without. The intrusion gains less (median overlap 0.55 → 0.60). Its dip stays too steep in most settings (median $64^\circ \rightarrow 61^\circ$, true 45°): three holes fix its susceptibility where they cross it, but not the slope between them.

Table 7. The spread of each measure over the 40 settings, without (A) and with the holes (C)

measure	truth	A: median	A: middle half	A: range	C: median	C: middle half	C: range
score	1.00	0.58	0.39–0.70	0.23–0.80	0.66	0.58–0.75	0.32–0.87
overlap: cube	1.00	0.63	0.33–0.77	0.12–0.87	0.74	0.64–0.81	0.27–0.87
overlap: intrusion	1.00	0.55	0.38–0.66	0.32–0.77	0.60	0.51–0.69	0.36–0.87
cube depth (m)	262	263	240–281	107–299	252	238–266	193–344
intrusion depth (m)	550	585	556–665	454–766	586	549–644	470–878
dip (°)	45	64	59–66	49–77	61	58–65	47–71
κ in the intrusion (SI)	1.00	0.71	0.64–0.79	0.33–0.97	0.73	0.58–0.80	0.35–0.95
settings scoring ≥ 0.7		10 / 40			18 / 40		
dip within 10° of the truth		5 / 40			7 / 40		
cube depth within 50 m		38 / 40			36 / 40		

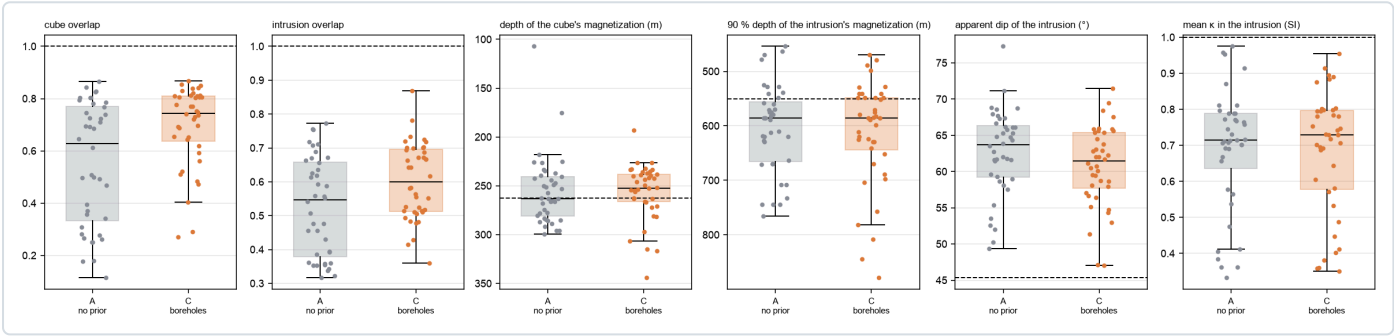


Figure 7 The measures of all 40 settings without (A) and with the holes (C); dashed: the true model measured the same way. Depth axes point down.

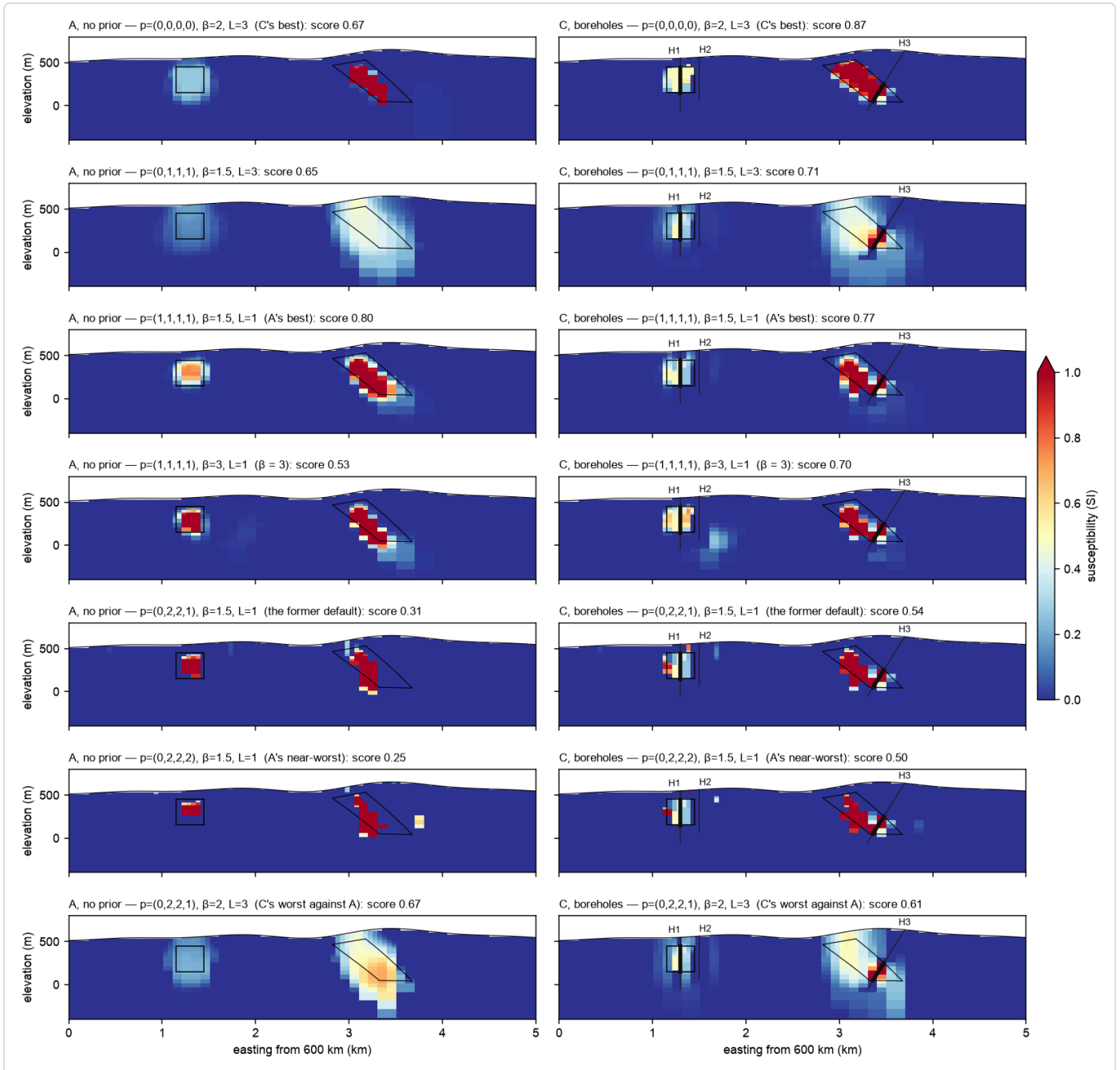


Figure 8 The same seven settings without (left) and with the holes (right): the holes' best, a blocky setting, the best without a prior, $\beta = 3$, GeoInv3D's former default, a setting that fails without a prior, and the setting where the holes lose most. Black outlines: the true bodies; thick black: the holes' magnetic intervals.

How far the holes reach. Comparing each setting with and without the holes (Figure 9a): at the holes' best setting the model changes by about 0.32 SI within 150 m of a hole, 0.20 SI at 150–250 m, 0.037 SI at 250–500 m and nothing beyond: the 150 m radius carries, through the regularization, to about 250 m. A test hole 224 m from the nearest hole, not given to the inversion, crosses the intrusion at 125–475 m. There the mean susceptibility is 0.98 SI with the holes and 1.16 SI without (true 1), and the intrusion's top and base are close to the truth only with the holes (Figure 9d). The holes and their 150 m reach take in 1.3 % of the cells of the area.

A caveat. Where the holes lose ($p(0,2,2,1) \cdot \beta 2 \cdot L 3$ and its neighbours), a likely cause is that the background intervals (0 SI) also reach 150 m: H3 and H5 run through background just above the intrusion, which may hold down its top. This was not checked setting by setting; a smaller radius for the background intervals would avoid it.

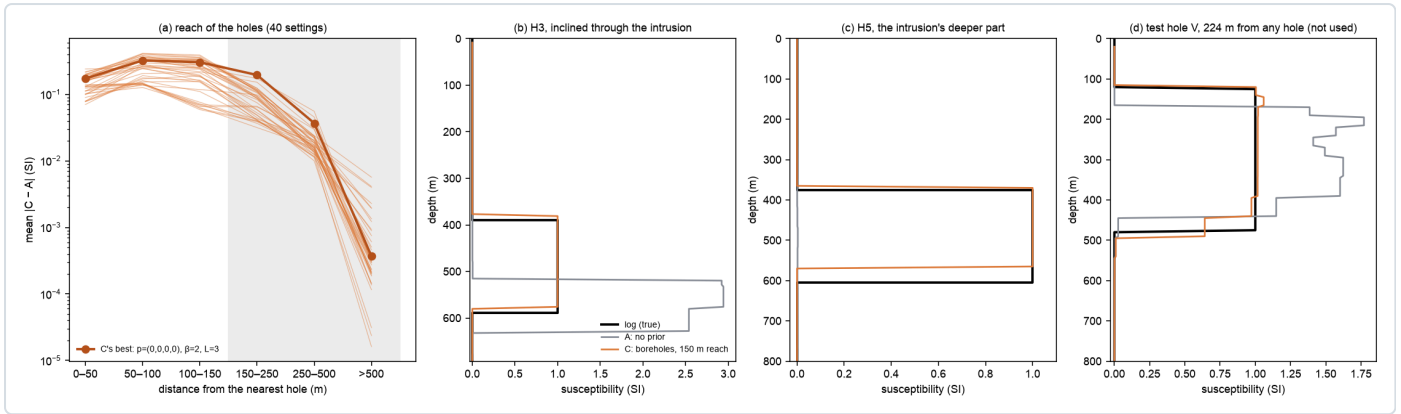


Figure 9 (a) The mean change $|C - A|$ with distance from the nearest hole (thin: every setting; thick: the holes' best setting); shaded: beyond the 150 m radius. (b)–(c) The susceptibility along two holes: the log (black), without (grey) and with the holes (orange), at the holes' best setting. (d) A test hole not given to the inversion.

8 The interpreted model as a reference

The interpretation is judged the same way, at its best and setting by setting (Figure 10):

- **Weight 1:** best 0.81, level with no prior (0.80). It beats no prior at 25 of 40 settings, by a median of only +0.02 (cube) and +0.02 (intrusion), mostly where no prior does badly. At the best setting without a prior it lowers the score from 0.80 to 0.63 (Figure 11). The data do change it: the interpreted cube, referenced at 0.75 SI, ends at 0.27 SI on average, the interpreted intrusion (0.5 SI) at 0.38 SI.
- **Weight 10:** best 0.70, median 0.46; better than no prior at 15 of 40 settings, the intrusion's overlap down by a median of 0.09. The interpreted bodies stay closer to their reference (0.43 and 0.46 SI).
- **Weight 100:** best 0.53, median 0.43; better at 10 of 40 settings. The interpreted bodies are held exactly at their reference (0.750 and 0.500 SI), as by a hard constraint, and the data are fitted elsewhere: at two settings with spurious bodies over 1 SI outside the cube (at the southern and northern edges of the area, 400–1,200 m deep), which leave the cube's overlap at 0.

Every one of these inversions fits the data as well as those without a prior (χ^2/N about 1). **An interpretation off by about 100 m and 10°, with susceptibilities 50 % off, did not improve the result.** Weakly weighted, the data undo most of it and it changes little; heavily weighted, it keeps magnetization where it was drawn, and nothing in the data fit shows it. A reference model is worth a heavy weight only if its geometry is better known than this; an accurate interpretation was not tested.

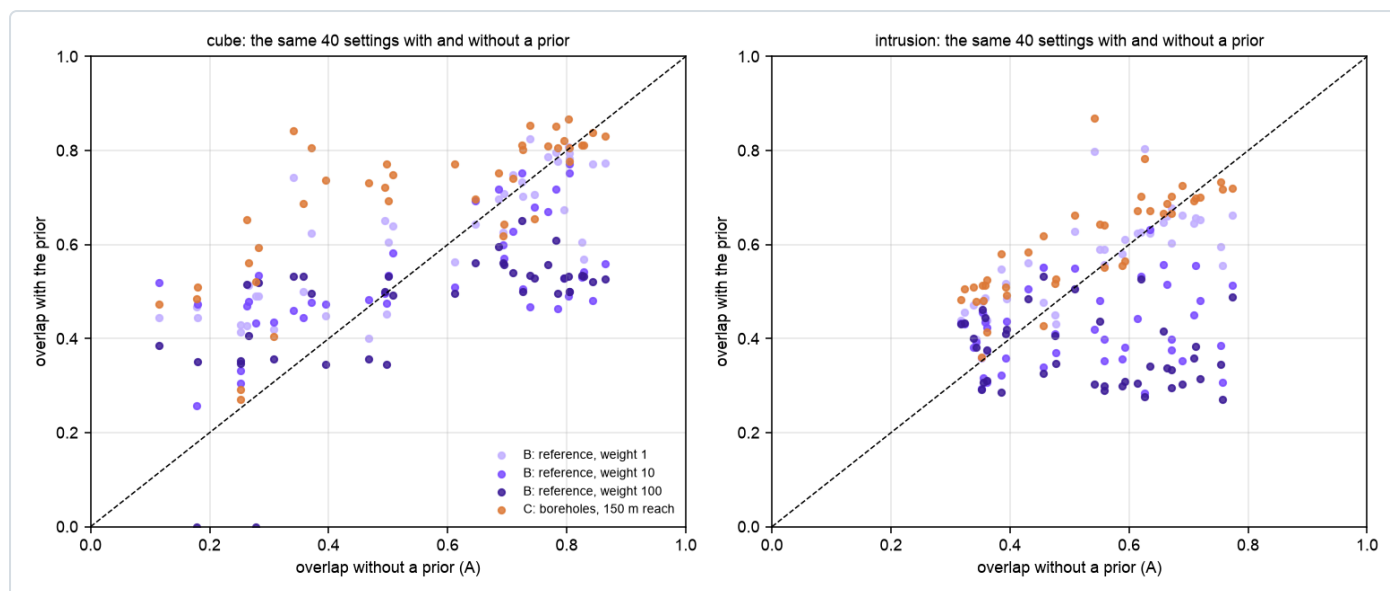


Figure 10 Each setting with (vertical) and without a prior (horizontal): the cube (left) and the intrusion (right). Above the dashed line the prior helps.

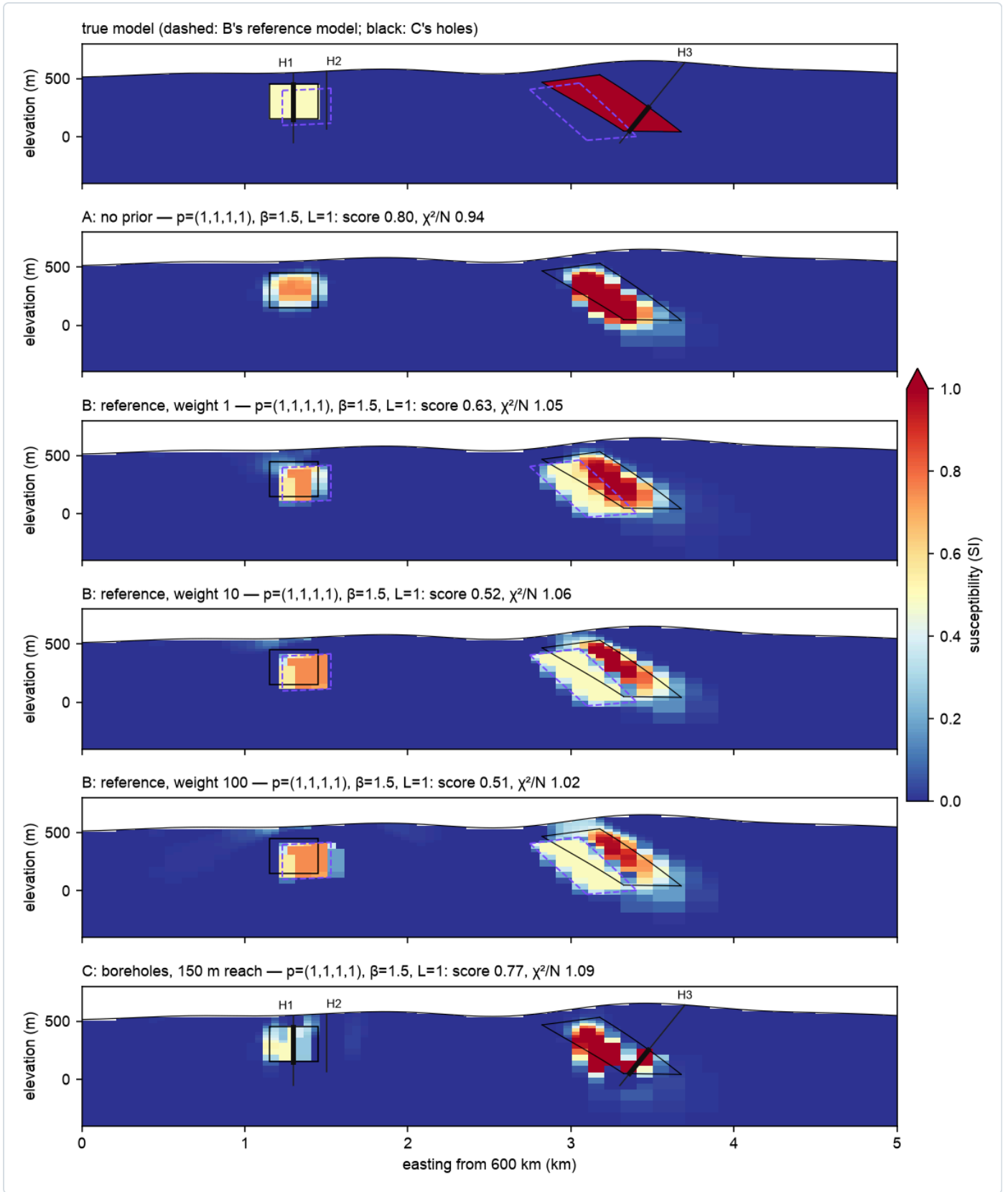


Figure 11 One setting (the best without a prior) in all five groups: only the prior differs.

9 Conclusions and recommendations

- The data fit cannot choose the settings or test a prior.** All 200 inversions have χ^2/N 0.85–1.12; their scores run from 0.22 to 0.87.
- Tune the L_p settings; it matters as much as the prior.** For compact, stubby bodies like these, $L = 3$, the norms $(1,1,1,1)$ and $\beta = 1.5$ –2 are the most robust: $p = (1,1,1,1)$, $\beta = 2$, $L = 3$ scores 0.75 without a prior and 0.78 with the

holes. GeoInv3D's former default ($p = (0,2,2,1)$, $L = 1$) scored 0.31 here; single magnetic inversions now default to $p = (1,1,1,1)$ at $L = 3$, which should still be checked on field data.

3. **Bring boreholes in, with a radius of influence.** They were the only prior that helped at most settings, most of all at the settings where the inversion alone fails. Give the background intervals a smaller radius than the magnetic ones, and let the page's borehole import set `radius_m` (now only in a JSON spec).
4. **Weight interpreted bodies lightly.** Where their position, dip or susceptibility is uncertain, a weight of 1 at most, and always compare with the inversion without them: a structure that appears only with the prior is not a result.
5. **On field data, choose settings with the holes.** The best settings found here need the true model and do not carry over directly. The patterns may: a smoother L , the norms $(1,1,1,1)$, β at most 2. Leaving one hole out at a time and checking how well the inversion predicts it is a test that needs no true model.

10 Limitations

- The best settings were chosen against the true model, which real surveys do not have.
- One model and one noise realization; induced magnetization only, no remanence.
- One interpretation error and one hole radius (150 m, the same for magnetic and background intervals) were tested.
- α_s , the cooling of the trade-off parameter and the IRLS thresholds were not varied; the data were thinned to 100 m.

Files

- Inputs and the parameters of the 200 inversions: `examples/output/synthetic_ablation_lp/inputs/` (`scripts/make_synthetic.py` ; `runs.json`)
- Runs: `deploy/ec2_sweep.py` (several EC2 instances) and `deploy/sweep_runner.py` (one machine); log `data/sweep.log`
- Results: `data/runs/` (all 200, not in the repository); the best of each group in `data/best/` ; every measure in `figures/numbers.json`
- Measures `scripts/metrics.py` , figures `scripts/make_figures.py` , this report `scripts/build_report.py`
- The boreholes' radius: `geoinv3d/methods/geology.py` (`radius_m`), described in `docs/geology_constraints.md`